



Maharshi Karve Stree Shikshan Samstha's
Cummins College of Engineering for Women, Pune
(An Autonomous Institute affiliated to Savitribai Phule Pune University)



PYTHON MINI-PROJECT

2024 - 25

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Branch / Year : SY Computer Science

Batch : B1

EMPLOYEE ATTRITION PREDICTOR

Problem Statement :

Employee attrition poses major challenges for organizations, including productivity loss and increased hiring costs. Traditional approaches to identifying at-risk employees are often reactive and lack data-driven support. This project aims to develop a web-based system that uses machine learning to predict employee attrition risk—categorized as low, moderate, or high—and provides graphical visualizations to help HR teams make informed, proactive decisions.

Keywords:

- Employee Attrition
- Machine Learning
- Random Forest Classifier
- Data Visualization
- Gradio Interface
- HR Analytics
- Python
- Data Preprocessing

Abstract :

- Employee turnover is increasing, creating a need for efficient prediction and prevention methods.
- This project uses IBM's HR Analytics dataset to train a machine learning model for predicting employee attrition.
- A **Random Forest Classifier** is chosen due to its effectiveness with both categorical and numerical data, and its ability to prevent overfitting.
- The model predicts whether an employee is likely to leave based on features such as:
Age, Gender, Department, Overtime status, Education field, Job role, Work-life balance.
- A **Gradio-based web interface** is developed, allowing HR staff to input employee data and receive immediate attrition risk predictions (low, moderate, high).
- The system includes data visualization tools like pie charts, histograms, scatter plots, and bar graphs to provide key HR insights.
- Designed to be **user-friendly**, the system requires minimal technical expertise, enabling seamless integration into HR operations.

Module-wise Description:

Data Collection Module:

- We collect the IBM HR Analytics dataset using the **Opendatasets** library directly from Kaggle.
- The dataset includes important fields like Age, Department, Job Role, OverTime, Education Field, etc.

Data Preprocessing Module:

- Categorical data such as Gender, Department, and OverTime are label-encoded and one-hot encoded.
- Feature scaling is done using **StandardScaler** to normalize the data.
- Irrelevant columns like EmployeeCount and Over18 are dropped.

Model Training Module:

- A **Random Forest Classifier** is trained using the processed dataset.
- The trained model is saved using **Pickle** to avoid retraining every time.

Prediction Module:

- Takes user input fields like Age, Gender, Department, etc.
- Encodes and scales the input based on the preprocessing pipeline.

- Loads the saved model using Pickle and predicts the attrition status.
- Classifies the result into risk levels (Low, Moderate, High) based on prediction probability.

Visualization Module:

- **Pie Charts:** Showing department-wise distribution of employees.
- **Histograms:** Displaying age distribution.
- **Scatter Plots:** Visualizing relationships like Monthly Income vs. Age.
- **Bar Graphs:** Showing average monthly income across different job roles.

Authentication Module:

- Login page with password verification.
- Password strength checking feature using a regex pattern ensuring a secure system.

Frontend Module (Web Interface):

- Built with **Gradio** to provide a simple and interactive web page.
- Allows login, data entry, prediction viewing, and graphical analysis through different pages.

Technology Selected and Technology Features Covered:

Python

Core programming language used for backend development and logic implementation.

Pandas & NumPy

Libraries used for efficient data handling, preprocessing, and numerical computations.

Scikit-Learn

Used to develop and evaluate the machine learning model (Random Forest Classifier).

Matplotlib

Used to create dynamic and interactive visualizations (e.g., pie charts, histograms).

Gradio

Provides a simple web interface for real-time prediction and data visualization.

Pickle

Used to save and load trained machine learning models efficiently.

Opendatasets

Automates downloading of datasets from Kaggle for training and evaluation.

Key References

- **Dataset:**
[IBM HR Analytics Employee Attrition & Performance](#)
- **Gradio Documentation:**
<https://gradio.app/docs>
- **Pandas Documentation:**
<https://pandas.pydata.org/docs>
- **Matplotlib Documentation:**
<https://matplotlib.org/stable/contents.html>
- **Scikit-Learn Documentation:**
<https://scikit-learn.org/stable/documentation.html>
- **Python Official Docs:**
<https://docs.python.org/3/>